

Reference excerpt 01-034

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and ECG waves, the time interval t_{aQP} between t_P Q and t_a is given by: $t_{aQP} = t_a - t_{QP} = IG / c$ (4) where t_a is the instant corresponding the a wave. The parameter t_{aQP} is the time interval between the instant when the pressure is maximum at point G (instant t_a) and the instant when the pressure is maximum at point RA (instant t_{QP}). METHODS The applicability of the relationship expressed in Eq. 4, can be tested by comparing the instantaneous velocity ($w(t)$) of the blood in the IJV calculated according to the equation shown above, with the instantaneous blood velocity ($w_D(t)$) measured with a Doppler scanner. However, the data and the results presented here are for illustration only and are not to be considered the result of a scientific study. Pressure waves velocity calculation The delay t_{aQP} is measured over the JVP+ECG trace as shown in Fig. 2. The distance IG is measured on the volunteers chest. The velocity c is the ratio between IG and t_{aQP} . Compliance calculation From the Moens-Korteweg equation we obtain the compliance per unit length: $C = CSA \times c^2$ (5) where CSA_x is the CSA of the IJV measured in G at the x wave. Pressure gradient calculation The pressure inside the IJV is calculated from the instantaneous CSA as follows: $p(t) = \frac{1}{C} CSA(t)$ (6) substituting the expression of $p(t)$ obtained above into Eq. (2) we obtain the expression for the pressure gradient: $p_z = \frac{1}{c} \frac{dCSA}{dt}$